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Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, “Surface-Energy Sign Classification for Timelike Kantowski–Sachs–Schwarzschild Junctions,” for publication in *General Relativity and Gravitation*. The manuscript identifies an invariant causal-sector split and explains geometrically why proposed black-hole-to-cosmology constructions placed inside the horizon are not subject to the exterior obstruction. Under the ordinary orientation, every compatible finite-rapidity junction from homogeneous Kantowski–Sachs geometry to the retained Schwarzschild $F > 0$ exterior requires negative Israel surface density. In the complementary $F < 0$ sector the ordering becomes sign-indefinite: paired future-infalling timelike witnesses in a regular chart realize opposite density signs for the same local bulk geometry and retained-side gluing, and an exact compatible choice has $p_s = \sigma/2$, strictly satisfying NEC, WEC, DEC, and the intrinsic shell SEC.

The Schwarzschild horizon is thereby identified as the invariant boundary between a universal negative-density exterior regime and an interior regime admitting fully energy-condition-satisfying local surface data. The manuscript does not claim to validate the dynamics of proposed completions or their generally spacelike transition layers. This combination of exact junction analysis, causal-sector classification, global branch control, and energy-condition analysis is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The manuscript source, figure-generation scripts, and verification suite are publicly available at github.com/RonBibb/junction-ec-paper.

Subject to the journal’s conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

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